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Cyclic voltammetric simultaneous determination of oxidizable amino acids using multivariate calibration methods

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Abstract

A voltammetric method for the simultaneous determination of oxidizable amino acids (cysteine, tyrosine and tryptophan) is proposed. The method is based on the analysis of multivariate data, which consists of cyclic voltammograms registered in the range from 400 to 1100 mV with a graphite–methacrylate composite electrode, by using multivariate calibration methods. Cyclic voltammograms of pure standards of these oxidizable amino acids show characteristic shapes. Thus, the resolution of mixtures of them by multivariate calibration lies in the differences in their voltammograms. The main difficulties encountered in these analyses are the high degree of overlapping and the lack of bilinearity of voltammetric data. The prediction abilities of principal component regression (PCR) and partial least squares regression (PLS) were assayed to the analysis of oxidizable amino acids in a series of synthetic samples. The quantification of cysteine and tryptophan was especially satisfactory since they are the most characteristic amino acids. ©2000 Elsevier Science B.V. All rights reserved.

Keywords: Oxidizable amino acids; Cyclic voltammetry; Graphite–methacrylate composite electrode; Multivariate calibration; Principal component regression; Partial least squares regression

1. Introduction

Amino acids are usually analyzed by liquid chromatographic methods [1], where a pre- or post-column derivatization reaction is applied in order to improve sensitivity. Ninhydrin, *o*-phthaldialdehyde, phenylisothiocyanate, dansyl chloride and 9-fluorenylmethylchloroformate are examples of typical reagents used for this purpose. However, the chromatographic separation is often complex, tedious and time-consuming. For this reason there is an interest in developing faster, simpler and cheaper procedures for amino acid analysis.

This paper describes a method for determining several oxidizable amino acids such as cysteine, tryptophan and tyrosine. These amino acids are important in the dietetic and pharmacological fields [2]. Tryptophan (Trp) is an essential amino acid for humans and herbivores, which is sometimes added to dietary and feed products as food fortifier. In addition, Trp is the limiting amino acid in various vegetable products. Cysteine (Cys) is not itself an essential amino acid, but is an additional source of sulfur for the organism. Tyrosine is not an essential amino acid either, but it is a direct precursor of various hormones, biogenic amines and neurotransmitters.

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In the literature, several electrochemical methods for determining oxidizable amino acids (in particular for cysteine analysis) have been proposed [3,4]. A big drawback is that other oxidizable compounds present in the samples may interfere with these analyses, which makes their practical applications limited.

Multivariate calibration techniques have become an alternative to the classical chromatographic methods mentioned above [5,6]. The chemometric approach is especially suitable for satisfying the great demand of amino acid determinations in many areas such as biochemistry, clinical chemistry, nutrition, agrochemistry and pharmaceutical products.

Multivariate calibration methods have usually been applied to the analysis of spectroscopic data, including UV–Visible spectrophotometry, circular dicroism, fluorometry, infrared and Raman spectroscopy, ESR, NMR, etc. [5,6]. Multivariate calibration methods have been used scarcely to the study of voltammetric data. The literature describes the determination of metal trace mixtures of thallium and lead by anodic stripping voltammetry, using multiple linear regression (MLR) [7] and partial least squares regression (PLS) [8,9]; arsenic [10] and mixtures of iron and titanium [11] by stripping potentiometry; mixtures of copper, cadmium and nickel by differential pulse polarography [12]; and other complex trace mixtures [13] and copper traces in wine [14] by differential pulse anodic stripping voltammetry and PLS. Some examples of the resolution of mixtures of organic compounds such as sulfonamides [15], anti-inflammatory agents [16], benzaldehyde [17], carbamate pesticides [18], nitrofuran derivatives [19] and pyrazine derivatives [20] by differential pulse polarography and PLS, and synthetic dye mixtures by voltammetry and PLS [21], have also been described.

In the present study, a voltammetric method for the simultaneous determination of oxidizable amino acids (Cys, Tyr and Trp) by multivariate calibration is proposed. Multivariate data to be analyzed consist of cyclic voltammograms registered in the range of potentials from 400 to 1100 mV by using a graphite–methacrylate working electrode. Cyclic voltammograms of pure standards of Cys, Tyr and Trp show characteristic shapes which enable these amino acids to be distinguished. Hence, the resolution of mixtures of these compounds by multivariate calibration lies in the differences in the voltammograms. However, these cyclic voltammograms were highly overlapped and no selective potential was found. In consequence, the determination of these amino acids by classical univariate calibration methods is strongly interfered. Another serious drawback of the voltammetric data is the lack of linearity and bilinearity (i.e. the shape of the voltammogram of the analyte was not unique). Modelling this type of system, especially non-bilinear data, is much more difficult, and so the resolution and quantification may be hindered.

The prediction abilities of several multivariate calibration methods such as PCR, PLS1 and PLS2, using a series of synthetic mixtures of Cys, Tyr and Trp, were investigated in this study.

2. Experimental

2.1. Reagents

All reagents were of analytical grade. Solutions were prepared using Millipore water. Amino acids were supplied by Merck. The background electrolyte solution was prepared from potassium chloride (Merck) and potassium monohydrogen phosphate (Merck). The pH of this solution was adjusted to 7.5 with a 37% (w/w) hydrochloric acid solution (Merck). Working electrodes were prepared by mixing a Sealer-Healer 1540 commercial methacrylate monomer (QH-57T-D, from Rohm and Haas), benzoyl peroxide (Fluka) as the initiator and graphite powder (Merck).

The composition of the synthetic samples in this study is shown in Table 1. Samples were prepared from three-analyte three-level factorial designs: for samples 1–24, concentrations were 6×10^{-5}, 4×10^{-5} and 2×10^{-5} M for Cys and Tyr and 4.8×10^{-5}, 3.2×10^{-5} and 1.6×10^{-5} M for Trp; for samples 25–36, concentrations were 4.8×10^{-5}, 4×10^{-5} and 3.2×10^{-5} M for Cys and Tyr and 3.8×10^{-5}, 3.2×10^{-5} and 2.6×10^{-5} M for Trp. All these samples were prepared by adding with a micropipette suitable volumes of 0.1 M stock solutions of each amino acid up to a final volume of 25 ml.

2.2. Apparatus

Cyclic voltammograms were obtained with a PGSTAT 20 Autolab potentiostat from Eco Chemie.

Table 1
Composition of the amino acid mixtures

Sample	Cys (M)	Trp (M)	Tyr (M)
1	6.0×10^{-5}	3.2×10^{-5}	4.0×10^{-5}
2	4.0×10^{-5}	4.8×10^{-5}	4.0×10^{-5}
3	4.0×10^{-5}	3.2×10^{-5}	6.0×10^{-5}
4	6.0×10^{-5}	4.8×10^{-5}	4.0×10^{-5}
5	6.0×10^{-5}	3.2×10^{-5}	6.0×10^{-5}
6	4.0×10^{-5}	4.8×10^{-5}	6.0×10^{-5}
7	6.0×10^{-5}	3.2×10^{-5}	2.0×10^{-5}
8	4.0×10^{-5}	1.6×10^{-5}	6.0×10^{-5}
9	2.0×10^{-5}	4.8×10^{-5}	4.0×10^{-5}
10	2.0×10^{-5}	3.2×10^{-5}	6.0×10^{-5}
11	4.0×10^{-5}	4.8×10^{-5}	2.0×10^{-5}
12	6.0×10^{-5}	1.6×10^{-5}	4.0×10^{-5}
13	6.0×10^{-5}	1.6×10^{-5}	2.0×10^{-5}
14	2.0×10^{-5}	4.8×10^{-5}	2.0×10^{-5}
15	2.0×10^{-5}	1.6×10^{-5}	6.0×10^{-5}
16	6.0×10^{-5}	4.8×10^{-5}	2.0×10^{-5}
17	6.0×10^{-5}	1.6×10^{-5}	6.0×10^{-5}
18	2.0×10^{-5}	4.8×10^{-5}	6.0×10^{-5}
19	4.0×10^{-5}	1.6×10^{-5}	2.0×10^{-5}
20	2.0×10^{-5}	3.2×10^{-5}	2.0×10^{-5}
21	2.0×10^{-5}	1.6×10^{-5}	4.0×10^{-5}
22	4.0×10^{-5}	3.2×10^{-5}	2.0×10^{-5}
23	4.0×10^{-5}	1.6×10^{-5}	4.0×10^{-5}
24	2.0×10^{-5}	3.2×10^{-5}	4.0×10^{-5}
25	3.2×10^{-5}	3.2×10^{-5}	4.8×10^{-5}
26	4.0×10^{-5}	2.6×10^{-5}	4.8×10^{-5}
27	4.8×10^{-5}	3.2×10^{-5}	3.2×10^{-5}
28	4.8×10^{-5}	2.6×10^{-5}	4.0×10^{-5}
29	3.2×10^{-5}	3.8×10^{-5}	4.0×10^{-5}
30	4.0×10^{-5}	3.8×10^{-5}	3.2×10^{-5}
31	4.8×10^{-5}	3.8×10^{-5}	3.2×10^{-5}
32	4.8×10^{-5}	2.6×10^{-5}	4.8×10^{-5}
33	3.2×10^{-5}	3.8×10^{-5}	4.8×10^{-5}
34	4.8×10^{-5}	2.6×10^{-5}	3.2×10^{-5}
35	3.2×10^{-5}	3.8×10^{-5}	3.2×10^{-5}
36	3.2×10^{-5}	2.6×10^{-5}	4.8×10^{-5}

The working electrode was home-made and consisted of a PVC electrode body (18 mm length and 6 mm i.d.) filled with graphite–methacrylate polymer, prepared according to ref. [22]. A double-junction reference electrode Ag/AgCl Orion 90-02-00, with the external chamber filled with a 0.1 M KCl solution, and a platinum auxiliary electrode were used. The pH of the support electrolyte solution was adjusted with a Crison micro-pH 2002 pH-meter using a combined Ingold glass electrode.

2.3. Data sets under study

The voltammetric data were registered from 400 to 1100 mV in steps of 25 mV with a scan speed of 50 mV/s under steady-state conditions. Experimentally, this was achieved after repeating 15 successive cycles, so that, for each sample, the 15th voltammogram was taken as multivariate data for further analysis. The total analysis time for a sample was about 7.5 min (0.5 min per scan). Samples from Table 1 were measured randomly.

A suitable strategy when working with small data sets (e.g. here a set of 36 sample mixtures) is predicting each sample using the remaining samples as standards for building the calibration model [23]. For each sample analysis, the dimension of the corresponding response data matrices were the number of potentials of each voltammogram n_p ($n_p = 58$ values of current intensities taken at a regular interval of 25 mV in the range 400–1100–400 mV) by the number of samples n_s ($n_s = 35$, for the calibration, $n_s = 1$ for prediction).

2.4. Multivariate calibration methods

For calculations MATLAB for Windows (Version 4.1) [24] was used. Multivariate calibration methods were from the PLS_Toolbox from B. Wise [25].

A common requirement for all first-order calibration methods is that unknown samples and standards must have the same nature, in order to model implicitly all contributions from the analytes, interferences and matrix effects present in the samples to the response. For this reason, samples for calibration and prediction have to be of similar nature and composition. Although multivariate calibration methods have been extensively described elsewhere [5,6,26], a brief description is given below:

Principal component regression (PCR) decomposes the experimental matrix of responses of the calibration set as follows:

$$\boldsymbol{R} = \boldsymbol{T}\boldsymbol{P}^{\mathbf{T}} + \boldsymbol{E} \quad (1)$$

$\boldsymbol{R}$ being the response matrix with a dimension $n_s \times n_p$ (number of standards by number of potentials), $\boldsymbol{T}$ the scores matrix ($n_s \times n_f$), $\boldsymbol{P}^{\mathbf{T}}$ the loading matrix ($n_f \times n_p$) (the superindex $\boldsymbol{T}$ indicates the transposed matrix) and $\boldsymbol{E}$ the residual error matrix

($n_s \times n_p$), n_f is the number of latent variables or factors included in the model which are able to keep the relevant information of data.

Next, the scores matrix $\boldsymbol{T}$ is correlated with the concentration matrix $\boldsymbol{C}$, using the expression:

$$\boldsymbol{C} = \boldsymbol{T}\boldsymbol{B} + \boldsymbol{E} \tag{2}$$

where $\boldsymbol{B}$ is the matrix of regression coefficients which is resolved by using a least squares procedure.

This model is subsequently applied to predict the concentrations of unknown samples from the equations:

$$\hat{\boldsymbol{T}}_{\text{unk}} = \boldsymbol{R}_{\text{unk}}\boldsymbol{P} \tag{3}$$

and, finally

$$\hat{\boldsymbol{C}}_{\text{unk}} = \hat{\boldsymbol{T}}_{\text{unk}}\boldsymbol{B} \tag{4}$$

where the subindex unk refers to the unknown samples.

Partial least squares regression (PLSR). The PLS algorithm takes simultaneously into account the information for the response matrix $\boldsymbol{R}$ (voltammetric data matrix) and the concentration matrix $\boldsymbol{C}$. In this way, factors from a PLSR model are calculated as those variables which describe the maximum amount of information of the concentration matrix as follows:

$$\boldsymbol{R} = \boldsymbol{T}\boldsymbol{P}^{\mathbf{T}} + \boldsymbol{E}' = \Sigma t_k p_k + \boldsymbol{E}' \tag{5}$$

$$\boldsymbol{C} = \boldsymbol{Q}\boldsymbol{S}^{\mathbf{T}} + \boldsymbol{E}'' = \Sigma q_k s_k + \boldsymbol{E}'' \tag{6}$$

where, $\boldsymbol{T}$ and $\boldsymbol{P}$ are the score and loading matrices associated with the response; $\boldsymbol{Q}$ and $\boldsymbol{S}$ the scores and loading of the concentration matrix; and $\boldsymbol{E}'$ and $\boldsymbol{E}''$ the unexplained information of responses and concentrations, respectively. The corresponding scores and loading for the kth factor are t_k, p_k, q_k and s_k.

The inner relationship between responses and concentrations in the PLS model is given from their corresponding scores factor by factor:

$$q_k = b_k t_k \tag{7}$$

where b_k represents the regression coefficients.

Scores and loading of each factor are not calculated at once but one after each other, in order to minimize the residuals in the concentration matrix using a mixed relationship which can be generalized for the kth factor as:

$$\boldsymbol{E}_k = \boldsymbol{E}_{k-1} - t_k b_k s_k^{\mathrm{T}} \tag{8}$$

where $\boldsymbol{E}_k$ and $\boldsymbol{E}_{k-1}$ are the residuals of concentrations not explained by a model with k and $k-1$ factors, respectively. Note that for the first factor $\boldsymbol{E}_0 = \boldsymbol{C}$.

This model can be used to predict the concentration of unknown samples, as in PCR.

Leave-one-out cross-validation was used to select the number of latent variables to be used in PCR and PLS. The optimum number of latent variables was that which minimized the PRESS function, calculated using the expression:

$$\text{PRESS}(k) = \sum_{i=1}^{\text{Samples}} (c_{i\,\text{true}} - \hat{c}_{i\,\text{calc}}(k))^2 \tag{9}$$

where k refers to the number of latent variables considered, $c_{i\,\text{true}}$ is the real concentration of analyte in the sample i and $\hat{c}_{i\,\text{calc}}(k)$ is the concentration calculated by multivariate calibration methods using k factors.

3. Results and discussion

A preliminary study was designed in order to find the experimental conditions which would provide reproducible measurements. Repeated voltammograms showed a decrease in the current intensity scan to scan and, after approximately 10 successive cycles a steady state was attained. From this point, the voltammograms were highly reproducible from cycle to cycle. Therefore, the voltammogram of each particular sample was taken after 15 cycles in order to ensure the reproducibility of data. The precision of measurements was studied with six replicates of a 10^{-4} M tryptophan solution. The relative standard deviation was 3%, which was sufficient for this kind of measurement.

Fig. 1 shows the cyclic voltammograms of Cys, Tyr and Trp in the range 400–1000 mV obtained with a graphite-methacrylate working electrode. Each compound showed a characteristic voltammogram whose shape was distinguishable from the others. However, these cyclic voltammograms were highly overlapped and no selective potential to determine such amino acids with classical univariate calibration was found. The similarity of voltammograms

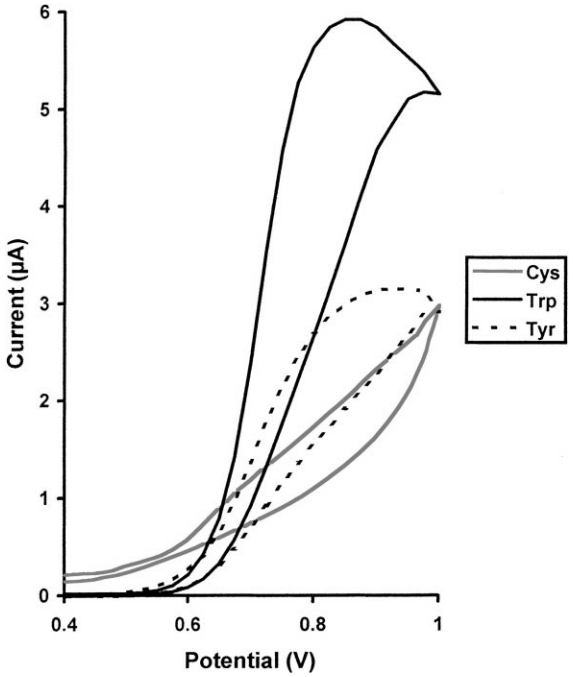


Fig. 1. Cyclic voltammograms of the oxidizable amino acids in 0.1 M KCl and phosphate (pH = 7.5) support electrolyte solution. Amino acid concentration 10^{-4} M.

was estimated using the correlation value (cosine value between couples of vectors, i.e. voltammograms in this case). Table 2 shows the values of correlation found in different cases. All cosine values reported in this Table were very close to 1, which indicated that voltammograms were highly similar in shape. The most dissimilar voltammograms were found between Cys and Tyr, while in the case of Tyr and Trp voltammograms were very similar. Correlation values (cosine) were also calculated for the oxidation and reduction ranges; there was no significant increase in dissimilarities and so the differences in the responses of each analyte were not stressed.

Table 2
Preliminary study of the similarity of the voltammograms of the amino acids from their correlation (cosine) values

Type of data	Cys–Trp	Cys–Tyr	Trp–Tyr
Whole range	0.9926	0.9860	0.9971
Whole range/background correction	0.9296	0.8948	0.9551
Oxidation range/background correction	0.9171	0.8379	0.9666
Reduction range/background correction	0.9644	0.9583	0.9571

The resolution and simultaneous determination of these amino acids using multivariate calibration methods lies on the particular features of the voltammetric response of each analyte (see Fig. 1). Consequently, since voltammograms were correlated closely, the quantification of such analytes was expected to be difficult.

The present study aims to establish the optimal working conditions and the best calibration methods for quantifying oxidizable amino acids. Thus, multivariate calibration techniques may be the methods of choice for determining a few analytes in a huge set of samples of similar nature and composition. For example, these characteristics are common in the industrial elaboration of feed samples, which generates samples candidates for analysis by multivariate calibration methods [27,28]. In this type of application, the best method of analysis comes from a suitable compromise between accuracy and speed.

A serious drawback found in the analysis of the voltammetric data was the lack of bilinearity. This problem is depicted in Fig. 2 for Trp, where a variation in the shape of the cyclic voltammograms depending on the concentration of amino acid is observed. Similar behaviour was encountered for Cys and Tyr, which did not have one single voltammogram shape either. Under these circumstances, the rank of each analyte was higher than one. Therefore, modelling non-bilinear data systems becomes much more difficult and, consequently, the resolution and quantification may be greatly hindered. In the present study, non-bilinearities were circumvented by limiting the amino acid concentration in the voltammetric cell to below 10^{-4} M. As can be seen in Fig. 2, working below that critical concentration level, the voltammograms seemed to be highly similar in shape and consequently, as an approximation, data could be considered bilinear.

The distribution of the samples on the first and second principal components from a principal component analysis of voltammetric data, showed that the first principal component was related to the maximum current intensity in the voltammograms, which directly depended on the total amount of amino acids in these samples. Samples with a high concentration of amino acids lie on the right, whereas samples with a low concentration of amino acids are placed to the left of the

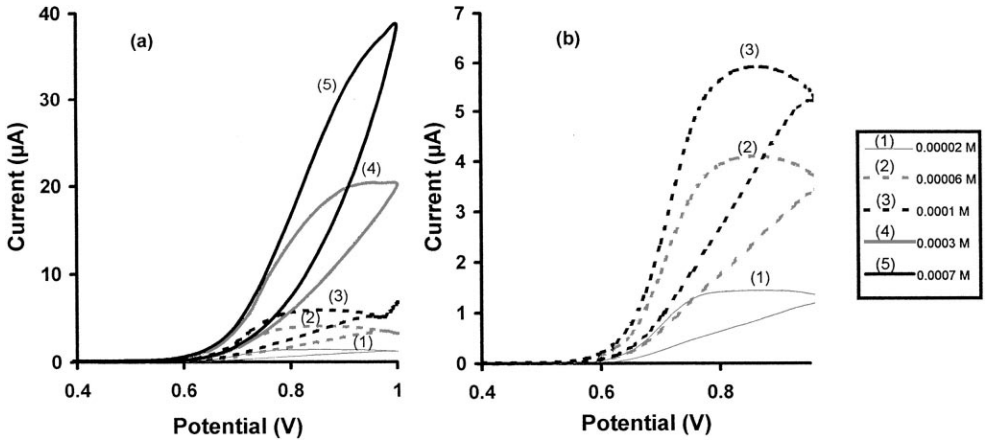


Fig. 2. Cyclic voltammograms of Trp solutions at different concentrations. (a) all concentrations; (b) concentrations below 10^{-4} M.

first PC axis. These two largest PC, which keep almost 98% of the variance in the response data, do not show any pattern for how samples with a high concentration of Cys, Tyr or Trp are distributed on these first and second PCs. Superior PCs cannot model the differences in the shape of voltammograms among amino acids, either.

As a general strategy, each sample was predicted using the remaining 35 samples as standards for building the model. The estimation of the number of factors was based on leave-one-out cross-validation by minimising the PRESS function (see Fig. 3). As a result, eight factors were found optimal for PCR, seven for PLS2, and six for each PLS1 model. A possible interpretation of these number of factors is given as follows: three factors were attributed to each analyte and an additional one to the background. Moreover, additional factors might be attributed to small shifts in the shapes of cyclic voltammograms and other possible interactions between analytes. These relatively high numbers of latent variables indicated, indeed, the complexity of the voltammetric data, since these systems cannot be modelled with only a few contributions.

PLS2 was used beforehand to check the optimum options for the analysis. Typical data pre-treatments such as mean-centering and autoscaling were investigated in order to check whether they were able to provide better predictions. Results concluded that determination became much worse. Similarly, derivative voltammograms were also investigated, but, again, the overall prediction error was much greater than with the original data (Table 3). Making blank corrections of voltammograms was another unsuccessful possibility checked. Besides, the analysis of data subsets such as the oxidative or reductive ranges were examined. No significant improvement in the predictions was observed over when the whole voltammetric range was used.

Table 3
Prediction error (%) in the determination of amino acids in the 36 synthetic mixtures using PCR, PLS1 and PLS2[a]

Calibration method	Cys		Trp		Tyr	
	PE (%)	NF[b]	PE (%)	NF	PE (%)	NF
PCR	11.4	8	10.5	8	18.5	8
PLS1	9.1	6	10.7	6	18.2	6
PLS2	11.0	7	10.2	7	20.3	7

[a] PE: prediction error.
[b] NF: number of factors.

Concentrations predicted with PCR, PLS1 and PLS2 methods were statistically compared with the actual values using *t*-tests of paired measurements. At a significant level $\alpha = 0.05$ and for 35 degrees of freedom, the corresponding t_{crit} value was larger than the t_{calc} in all cases, so that there were no significant differences in the real and calculated amino acid concentrations. In general, PLS1 seemed to be the most accurate quantification technique (with the exception of Trp) while PCR gave slightly higher prediction error. However, these differences were not statistically significant. Cys and Trp could be quantified with

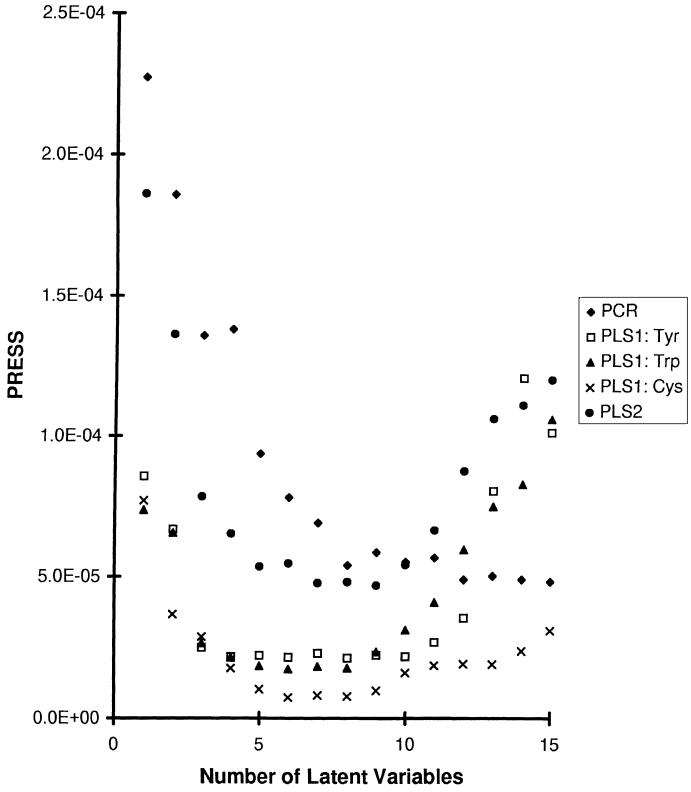


Fig. 3. Variation of PRESS vs. the number of latent variables from leave-one-out cross-validation analysis of the 36-samples set for PCR, PLS1 and PLS2 methods.

prediction errors ranging around 10%, while Tyr had poorer predictions.

4. Conclusions

This paper presents an initial study of the possibilities of voltammetric techniques in the simultaneous determination of mixtures of oxidizable amino acids. Cyclic voltammograms of mixtures of Cys, Trp and Tyr can be used as multivariate data for the determination of such amino acids. Hence, these oxidizable amino acids can be quantified in complex mixtures from a simple recording of the voltammogram. The accuracy in these determinations is about 10% for Cys and Trp and 18% for Tyr.

The method proposed here can serve as a basis for further analysis of these amino acids in industrial feed samples which contain other amino acids and complex matrix compositions, in which the accuracy of the analysis is not as critical as the speed in obtaining the results. In this way, multivariate calibration methods become an attractive alternative to chromatographic techniques, which are more accurate but also more tedious and time-consuming. Moreover, the electrochemistry of the process can be improved by using potential step methods to enhance the peak resolution and the electrode can be modified with electrocatalyzers to increase the selectivity of each amino acid towards others and the sample matrix. Further work is being done in these directions.

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